The Name of the Title Is Hope

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Abstract

LLM agents with web access are increasingly becoming the default for user-facing question answering. Reliable evaluation in this setting requires tests that remain unseen at train time and cannot be answered via verbatim lookup at inference. However, standard practice still relies on static benchmarks that age quickly and are vulnerable to contamination. We analyze two channels: pretraining leakage, where test items appear in the pretraining data, and retrieval-time leakage, where the system surfaces the exact target answer verbatim during evaluation and reports it rather than demonstrating genuine problem solving. We uncover evidence across five QA datasets for both channels and show that static evaluation can overestimate capability. We introduce DynamicWebQA, a dynamic evaluation framework for web-RAG that constructs questions and verifiable answers at evaluation time from current web sources. The framework enforces multi-document grounding, records evidence chains, and offers controllable complexity via source-set size, number of hops, and reasoning constraints. By generating fresh test instances, DynamicWebQA reduces memorization risk, mitigates retrieval-time leakage, and naturally covers time-sensitive queries.

CCS Concepts

• **Do Not Use This Code → Generate the Correct Terms for Your Paper**; *Generate the Correct Terms for Your Paper*; Generate the Correct Terms for Your Paper; Generate the Correct Terms for Your Paper.

Keywords

Do, Not, Use, This, Code, Put, the, Correct, Terms, for, Your, Paper

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1 Introduction

Benchmark leakage refers to the contamination that occurs when a model has prior exposure to a benchmark's test items during pretraining, leading to inflated evaluation scores and undermining the benchmark's validity [43, 44]. Since most LLMs are pretrained indiscriminately on publicly available web content, there is a high likelihood that they might have encountered benchmark datasets hosted on platforms like HuggingFace, GitHub, Kaggle, and similar sites. If the base LLM used in a WebQA agent has already seen the test examples during pretraining, the evaluation effectively becomes a case of testing on the training set — severely compromising the credibility of the reported results.

Dynamic benchmarking is an emerging evaluation paradigm that generates or adapts test samples at evaluation time, so models are scored on fresh, unseen data and testset contamination is avoided [18, 25, 33, 42, 44, 45]. That means the agent does not see the same test sample twice, effectively addressing issues such as LLM memorization.

In this work, we refer to information that can change over time, such as the stock price of Tesla, as evolving information. Arguably, this type of content could be labeled as dynamic content or temporal content, but that might cause confusion, since we are already using the term "dynamic" to describe an evaluation paradigm (dynamic benchmarking), and "temporal" might lead readers to associate it with time-series data. To avoid such confusion, we adopt the term evolving information to specifically denote factual content that may shift over time without implying a particular format or evaluation framework.

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2 Related Work

Open-Domain QA (ODQA). The ODQA task studies how to find and use evidence from large corpora to answer free-form questions. A sequence of datasets marked key shifts in task design: SQuAD [31] established span-based reading comprehension; Natural Questions [19] moved to real search queries; TriviaQA [15] stressed compositional, trivia-style questions with substantial lexical and syntactic variation; and HotpotQA [40] formalized multi-hop reasoning across multiple documents with sentence-level supporting facts and comparison questions, while including distractors to test focus under noise. Models such as fusion-in-decoder (FiD) [13] showed that aggregating many retrieved passages is a powerful way to synthesize scattered evidence.

Another important dimension in ODQA is *time*. RealTime QA [17] and FreshQA [37] evaluate whether systems can answer what is true *right now*, revealing failure modes when indices are stale or retrieved content is outdated or contradictory. These settings highlight needs that go beyond span extraction: recognizing when the answer has changed, when a question carries a false premise, and when recent evidence is required over older but higher-ranked documents. Notably, surface simplicity does not imply task simplicity. SimpleQA [38] demonstrates that short, seemingly straightforward questions can still require precise world knowledge or multi-step reasoning, indicating that difficulty is not well represented by length or phrasing.

Early ODQA followed a retrieve-then-read paradigm with extractive readers over IR-retrieved text. DrQA used TF-IDF/BM25 over Wikipedia coupled with a neural span extractor [6]. BERTserini then paired BM25 (Anserini) with a BERT reader to push end-to-end accuracy [39]. To support multi-hop questions, subsequent work leveraged structured signals: Asai et al. learned to retrieve reasoning paths over the Wikipedia hyperlink graph to connect supporting pages [2]; in parallel, document–entity heterogeneous graphs (e.g., GraftNet) aggregated evidence across passages and entities for open-domain QA [34]. Dense retrieval reduced lexical mismatch: ORQA learned a latent retriever from QA pairs (no explicit IR supervision) [21], and DPR's dual-encoder with supervised negatives yielded large gains over strong BM25 baselines [16]. These systems still relied on static indices, motivating later generation-conditioned approaches (RAG) for better robustness to paraphrase and freshness.

Retrieval-Augmented Generation (RAG). Modern RAG approaches follow a common pipeline – given a query, retrieve top-k passages from a non-parametric memory and condition a generator on them. Foundational formulations include RAG [22], REALM [11], RETRO [3], and ATLAS [14], which together demonstrate that query-time retrieval improves knowledge-intensive tasks while enabling the external memory to be updated without re-training the parametric model. Recent surveys study components and variants of this paradigm [10].

In the literature, RAG is distinguished from broader tool-augmented generation – RAG requires *query-conditioned retrieval* whose contents directly condition the generator. Calculators, code execution, or static always-prepended context are typically treated as non-retrieval tools [26]. In practice, the retrieval index can be the open web or enterprise/local corpora, and the mechanism can be lexical (e.g., BM25) or dense; the generator then aggregates evidence across multiple passages [13, 22].

Beyond single-shot RAG, agentic frameworks let models iteratively issue searches, follow links, and cite sources. WebGPT augments an LLM with browsing and requires collecting references during answering [28]. ReAct further couples reasoning traces with tool calls (e.g., search, calculator) in a unified prompting loop, enabling multi-hop planning and verification [41]. These approaches often embed RAG steps inside a broader plan-act-observe loop, allowing complex queries that require cross-document synthesis.

Data Contamination. Data contamination arises when evaluation items (or close paraphrases and answer-bearing spans) leak into the corpora used for pretraining, instruction tuning, or fine-tuning. In such cases, test behaviour can devolve into *lookup* of memorized strings rather than the intended inference or problem solving, inflating reported performance and obscuring brittleness [23, 32]. Prior studies connect leakage to memorization dynamics: larger models memorize more and earlier in training; duplication and longer prompts amplify verbatim regurgitation; and model capacity allows substantial unintended storage of datapoint-specific information [5, 27, 36]. At the same time, verbatim completion is not definitive evidence of membership – models can perfectly complete sequences never explicitly seen during training – implying membership tests must be interpreted cautiously [24]. The measurable impact of contamination is further mediated by dataset construction, model scale, and properties of pretraining data [23].

Ravaut et al. [32] organize detection techniques by *data availability*. With candidate corpora available (*open-data*), methods compare evaluation items against sources via instance-/dataset-level span matching, embedding similarity, or paraphrase detection. When training data are unavailable (*closed-data*), researchers rely on indirect evidence such as performance analyses, membership-inference attacks, and instance-level probes of memorization and model confidence. Despite blind spots, approaches based on span matching remain the practical default in most foundational efforts [4, 7, 9, 20, 29, 30] – as they are auditable and more importantly, tractable at web scale. Auditing a single question or sentence against a large crawl is a needle-in-a-haystack search; doing so for entire benchmarks across billions of documents renders more sophisticated but compute-heavy detectors infeasible. Moreover, behavioural signals alone rarely certify contamination [23, 24].

Retrieval-augmented and agentic QA introduce a new concern – these systems may retrieve and quote target items from web or enterprise indices at *run time*, creating search-time data contamination that boosts scores without demonstrating reasoning [12]. Han et al. [12] show that even small leakage rates (e.g., on the order of a few percent) can reorder competitive leaderboards once online access is enabled, undermining trust relative to purely offline evaluation.

Dynamic Benchmarking. Most benchmarks for agentic or retrieval-augmented QA remain static and publicly accessible (e.g., HotpotQA), which makes them susceptible to contamination and memorization [5, 23, 27, 32, 40]. As models and training corpora grow, leakage can inflate reported scores on static benchmarks. This has motivated *dynamic* evaluation paradigms that are more robust towards contamination and memorization [8, 33, 42, 46].

Early dynamic platforms such as Dynabench and Dynaboard introduced human- and model-in-the-loop data collection [18, 25, 35], but their reliance on crowdsourcing limits scalability. More recent approaches leverage domain-specific settings (e.g., AndroidWorld for interactive agents [33]) or graph-based generation to create evaluation samples with controllable complexity and cross-round consistency [8, 42, 46]. While graph-anchored methods have proven effective in tasks like reasoning and KGQA [8, 42], their application to ODQA is still limited, a gap we address in this work.

3 Modeling Contamination in RAG-Based QA

Retrieval-Augmented Generation (RAG) systems combine a parametric model M with external sources retrieved at inference. This creates two leakage channels: (i) *pretraining contamination*, where an evaluation item (or a close paraphrase / answer-bearing span) appears in the corpus used to pretrain M; and (ii) *retrieval-time contamination*, where the retriever surfaces such material during evaluation, enabling lookup rather than reasoning. Our experiments instantiate web-enabled RAG ("web-RAG"), but the formalism below applies to RAG broadly.

3.1 Definitions

Let M be pretrained on corpus D_M. Let D_{eval} contain items $x = (q, a)$ with question q and answer a. Let $\mathcal{R}(q)$ be the set of documents returned by the retriever for query q. Following Ravaut et al. [32], let f be a similarity function and τ a threshold; τ=1 corresponds to *strict/verbatim* contamination.

We study both *question-only* and *question+answer* leakage, noting that even input-only (question-only) leakage can shift measured performance [23]. We consider two views of an item:

$$\pi_{\text{in}}(q, a) = q \qquad \text{(\textit{question-only}, a.k.a. input-only)},$$

$$\pi_{\text{in+lab}}(q, a) = q \,\|\, a \qquad \text{(\textit{question+answer}, a.k.a. input+label)}.$$

Contamination criterion. For a channel $C \in \{\text{pre}, \text{ret}\}$ and view $v \in \{\text{in}, \text{in+lab}\}$, we say D_{eval} is contaminated if

$$\exists\, x \in D_{\text{eval}},\ \exists\, d \in \mathcal{S}_C(x) \ \text{ s.t. } \ f(\pi_v(x), d) \geq \tau, \tag{1}$$

with source sets

$$\mathcal{S}_{\text{pre}}(x) = D_M, \qquad \mathcal{S}_{\text{ret}}(x) = \mathcal{R}(q).$$

When evidence must be explicitly *answer-bearing*, we additionally require a predicate $g(a, d) \in \{0, 1\}$ (e.g., answer mention or high-overlap), and record positives of

$$\exists\, x, d \ \text{ s.t. } \ f(q, d) \geq \tau_q \ \wedge \ g(a, d) = 1. \tag{2}$$

3.2 Implementation details

We instantiate f using five auditable open-data detectors frequently used in LLM pre-training studies and consolidated in [32]:

- **Exact Match (EM)** [9].
- **8-gram words** (GPT-2 pre-training study) [30].
- **13-gram words** (GPT-3 pre-training study) [4].
- **50-character substring** (GPT-4 technical report) [29].
- **8-gram words with 70% threshold** (PaLM pre-training study) [7].

Span-based detectors scale to long documents, handle the length mismatch between short questions and web pages, and yield concrete evidence spans. While semantic detectors (embeddings / paraphrase) may increase recall, they are harder to audit and become infeasible across disparate lengths; we therefore adopt span-based methods as primary signals, consistent with cautions in [24] and common audit practice [32].

Pretraining channel (candidate retrieval). Full-membership testing against all datapoints in web-scale pretraining corpora is infeasible. Following the candidate-set approximation in open-data audits [23], we index the CommonCrawl C4 corpus in an Elastic-compatible service (AWS OpenSearch). For each $x = (q, a)$, we retrieve top-k candidates with BM25 (default k=1000) and apply the detectors between π_{in} or $\pi_{\text{in+lab}}$ and each candidate document. Any positive flag indicates potential pretraining contamination.

Retrieval-time channel (web-RAG).. For each q, we query SearxNG [1] with the raw question string, pool results from the first three result pages across underlying engines, and define $\mathcal{R}(q)$ accordingly. We (i) compute contamination using (1) and, when needed, (2) over this pool, and (ii) use the same pool as retrieval context for answer generation. This isolates run-time leakage introduced by search from memorization in M.

Reporting. For each dataset and channel we provide: (a) question-only and question+answer contamination rates; (b) per-detector breakdowns; and (c) overlap between pretraining and retrieval-time hits.

4 Method

We propose a method that leverages *seed graphs* to anchor future question generation, avoiding reliance on static datapoints that are vulnerable to memorization or data contamination. Each evaluation round produces a fresh QA pair from every graph, ensuring that no fixed test items can be leaked in advance while maintaining semantic comparability across rounds through the shared anchors.

4.1 Seed Graphs

At evaluation round t, we maintain a set $\mathcal{S}_t = \{G_1^{(t)}, \ldots, G_{n_t}^{(t)}\}$ of seed graphs. Each graph encodes the evidence trail (e.g., queries, retrieved documents, answer node) for a distinct information need. To support evolving information, graphs are constructed at evaluation time.

4.2 QA Generation

The generation function Gen takes a seed graph $G_i^{(t)}$, a random seed $r_{i,t} \sim \mathcal{R}$, and a fixed generation configuration θ to produce one QA pair:

$$(q_{i,t}, a_{i,t}) = \text{Gen}(G_i^{(t)}; r_{i,t}, \theta) \tag{3}$$

Here, θ denotes the fixed generation configuration (frozen prompt template and decoding settings, e.g., temperature, top-p, stop tokens). Gen is an LLM-based generator. $r_{i,t}$ is an i.i.d. sample from the randomness source $\mathcal{R}$, introducing per-sample stochasticity. Holding θ constant and varying only $r_{i,t}$ ensures that each system sees independent but identically distributed QA samples.

4.3 Sampling and Distributional Properties

Sampling an independent seed $r_{i,t} \sim \mathcal{R}$ for each graph yields the per-graph distribution:

$$\mathcal{P}_{i,t} = \mathrm{Dist}(\mathrm{Gen}(G_i^{(t)}; r, \theta)) \tag{4}$$

Because seeds are independent across graphs, the round-t dataset $D_t = \{(q_{1,t}, a_{1,t}), \ldots, (q_{n_t,t}, a_{n_t,t})\}$ is an i.i.d. sample from the product measure:

$$\mathcal{P}_t = \prod_{i=1}^{n_t} \mathcal{P}_{i,t} \tag{5}$$

For a model M under evaluation, let $\mathrm{score}((q, a), M)$ be any per-question metric (e.g., exact match, F1, log-loss). The aggregate dataset score is defined as:

$$\mathrm{Agg}(D_t, M) = \frac{1}{n_t} \sum_{i=1}^{n_t} \mathrm{score}((q_{i,t}, a_{i,t}), M) \tag{6}$$

By linearity of expectation and independence of $(q_{i,t}, a_{i,t}) \sim \mathcal{P}_{i,t}$:

$$\mathbb{E}_{D_t \sim \mathcal{P}_t}[\mathrm{Agg}(D_t, M)] = \frac{1}{n_t} \sum_{i=1}^{n_t} \mathbb{E}_{(q,a) \sim \mathcal{P}_{i,t}}[\mathrm{score}((q, a), M)] \tag{7}$$

This shows that the expected accuracy is simply the average of the per-graph expectations—so every seed graph contributes equally.

4.4 Collision Bound

Assume for every seed graph $G_i^{(t)}$, the generator can produce K distinct question–answer pairs:

$$C_{i,t} = \{(q, a) \in \mathcal{P}_{i,t} \mid \Pr((q, a)) > 0\}, \quad |C_{i,t}| = K$$

To characterize overlap across rounds $u < v$, define:

$$J_{i,u,v} = |C_{i,u} \cap C_{i,v}|, \quad J_{\max} = \max_{u<v} J_{i,u,v}$$

Because answers and evidence evolve over time, $J_{\max}$ tends to remain small for fast-changing topics.

Collision Probability. Sampling one QA pair per round, the probability that seed graph i produces the same QA pair twice within t rounds satisfies:

$$\Pr[\text{collision within } t] \le \frac{t(t-1)}{2K^2} J_{\max} \tag{8}$$

Collision Control. To guarantee that this probability is at most $\delta \in (0, 1)$, it suffices to choose K such that:

$$K \ge \sqrt{\frac{t(t-1)}{2\delta} J_{\max}} \tag{9}$$

Equation (7) captures key trade-offs:

- Larger K (richer candidate pool) reduces collision probability for fixed t.
- Fewer evaluation rounds t allow a smaller K for the same error tolerance δ.
- Smaller $J_{\max}$ (i.e., greater semantic drift) reduces required K.

Because each seed graph is rebuilt from fresh evidence each round, its candidate set changes, helping keep $J_{\max}$ small. Equation (6) quantifies the collision risk, while Equation (7) provides a tunable design guideline for selecting K.

4.5 Cross-Round Reporting and Compatibility

Because each round rebuilds the seed graph set $\mathcal{S}_t$, the underlying distribution $\mathcal{P}_t = \prod_i \mathcal{P}_{i,t}$ may drift over time, especially for evolving topics. Consequently, raw accuracies from different rounds may not be directly comparable. We offer two strategies:

i) Snapshot Evaluation. Fix a reference round t^* and freeze its seed graphs $\mathcal{S}_{t^*}$. All systems are then evaluated on the same dataset D_{t^*}, ensuring identical test conditions.

ii) Macro-Averaged Score. For longitudinal tracking, aggregate performance across a window $\{t_1, \ldots, t_T\}$ (e.g., most recent T rounds) using:

$$\mathrm{Score}_{\mathrm{macro}} = \frac{1}{T} \sum_{j=1}^{T} \mathrm{Agg}(D_{t_j}, M) \tag{10}$$

Each D_{t_j} is i.i.d. from its own $\mathcal{P}_{t_j}$, so Eq. (8) summarizes average effectiveness under the benchmark's natural evolution, without letting any single round dominate.

4.6 Verifying Distributional Consistency

If the generation is stable, items produced in different rounds should be indistinguishable once mapped into a semantics–preserving space. Sentence embeddings place short texts so that nearby points share topical content; consequently, a shift in topics should manifest as a change in the *distribution of pairwise distances*—not just a mean shift.

We concatenate each question–answer pair and compute a sentence embedding with `intfloat/e5-base-v2`. Vectors are ℓ_2–normalized so Euclidean and cosine distances are monotone-equivalent.

On the Euclidean distance matrix we run a K-sample omnibus test with PERMANOVA (10k label permutations; report pseudo-F, p, and R^2). To localize potential differences, we run the Energy two-sample test for all $\binom{K}{2}$ round pairs (10k permutations; Benjamini–Hochberg FDR at α=0.05). We add two descriptive visuals: a 2D PCA of embeddings colored by round, and a *centered mean-distance* map that removes within-round dispersion; this quantity is proportional to the biased energy distance and serves as an effect-size view.

5 Results

5.1 Contamination Modeling in QA Datasets

5.2 Model Performance

5.3 Distributional Consistency: Results

Omnibus. PERMANOVA detects no round effect in either split (Table 1). With 10,000 permutations the p-value resolution is 10^{-4}; both splits yield $p \ge 0.99$ and $R^2 \le 0.134\%$, indicating negligible variance explained by "round".

Pairwise. For the 45 round pairs per split, no comparison is significant after BH–FDR (α=0.05). E-statistics are tiny (10^{-4}–10^{-3}) and centered near zero (Table 2); maxima are $< 10^{-3}$.

Visual evidence. PCA projections (Fig. 5) show full intermixing by round (PC1/PC2 <4% explained each). Drift heatmaps of signed E-statistics (Fig. 6) concentrate near zero without structure. Centered mean-distance maps and relative deviations (Fig. 7) show max absolute effects $\le 1.5{\times}10^{-3}$, i.e., <0.25% of the within-round scale.

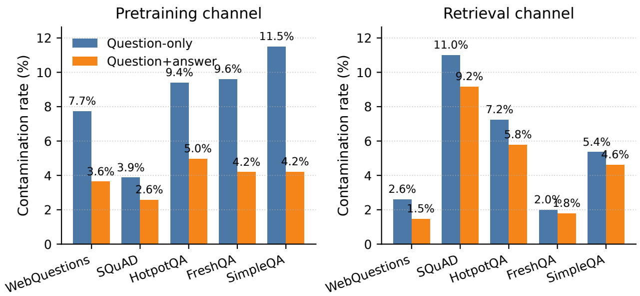

Figure 1: Contamination rates by dataset across the two leakage channels defined in the modeling framework: (a) pretraining contamination (source set $\mathcal{S}_{\text{pre}}=D_M$), and (b) retrieval channel contamination (source set $\mathcal{S}_{\text{ret}}=\mathcal{R}(q)$).

Table 1: Omnibus PERMANOVA across K=10 rounds (Euclidean on ℓ_2-normalized embeddings; 10k permutations).

Split	N	Pseudo-F	R^2 (%)	Perms	p-value
Highly Contaminated	5,000	0.7429	0.1338	10,000	≥ 0.99
Randomly Sampled	4,996	0.6476	0.1168	10,000	≥ 0.99

Table 2: Summary of pairwise Energy tests (signed, U-centered E-statistic).

Split	#Sig (FDR)	Mean	Median	
Highly Contaminated	0/45	-5.14×10^{-4}	-6.31×10^{-4}	3.16
Randomly Sampled	0/45	-6.58×10^{-4}	-7.10×10^{-4}	2.08

6 Conclusion

TBD

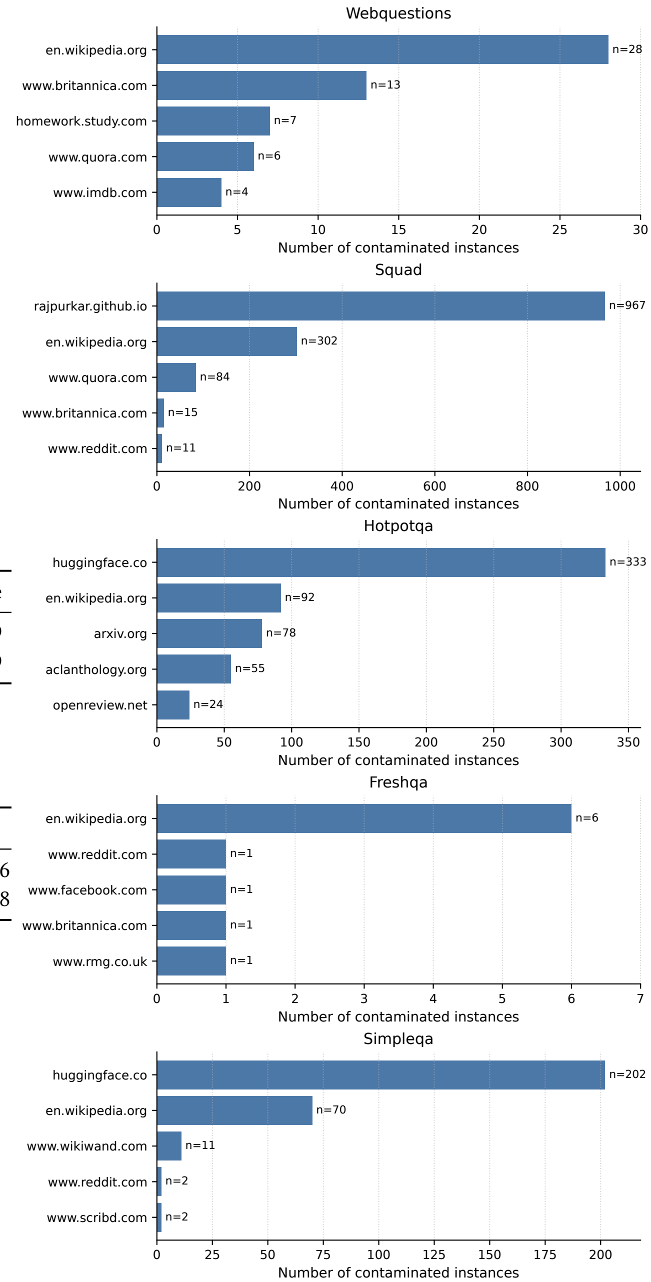

Figure 2: Top contamination sources during retrieval for each dataset. For each dataset, we show the top-5 domains by number of contaminated instances.

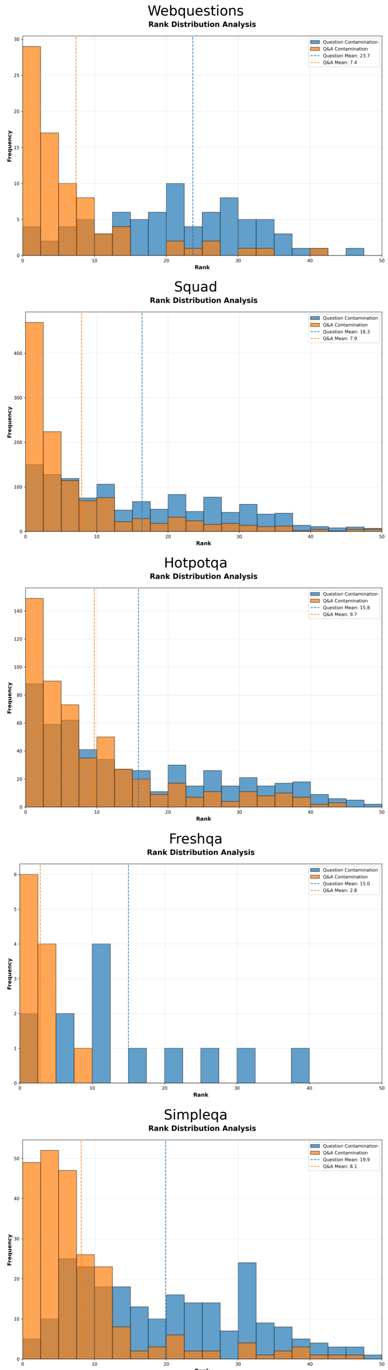

Figure 3: Rank distribution of contaminated evidence surfaced by the retriever, shown per dataset. Histograms separately show question-only and question+answer contamination where available.

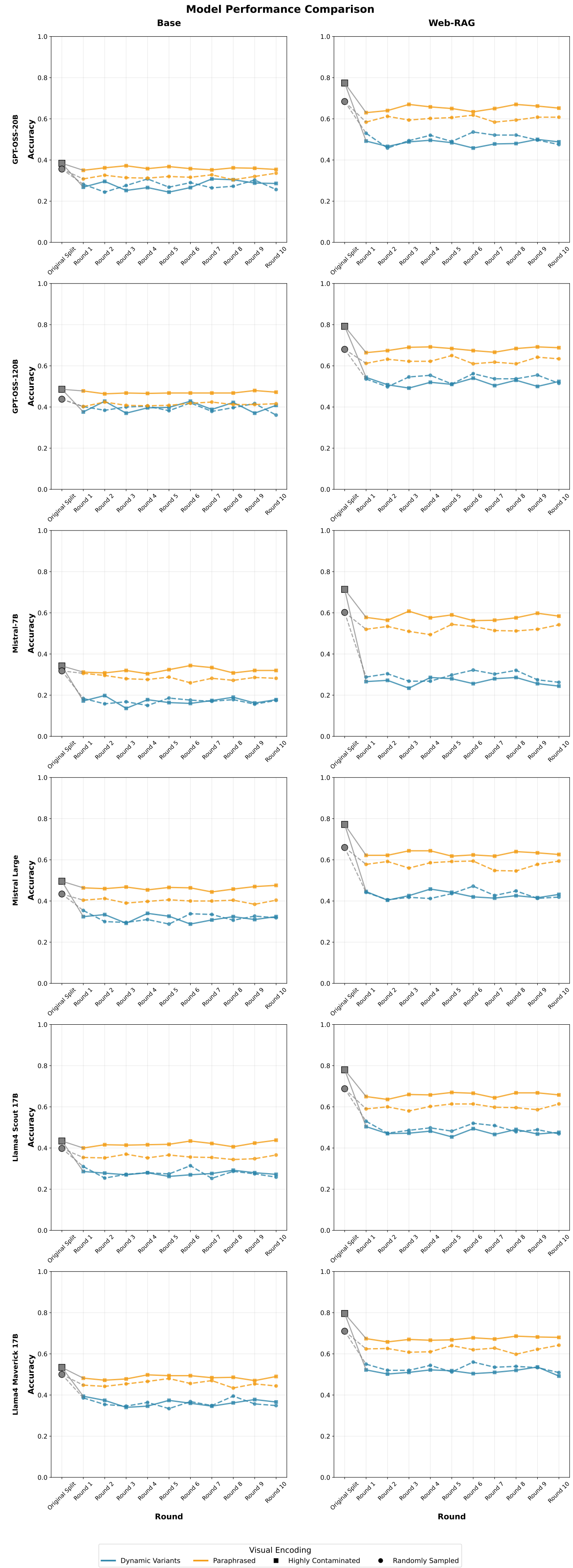

Figure 4: Caption TBD

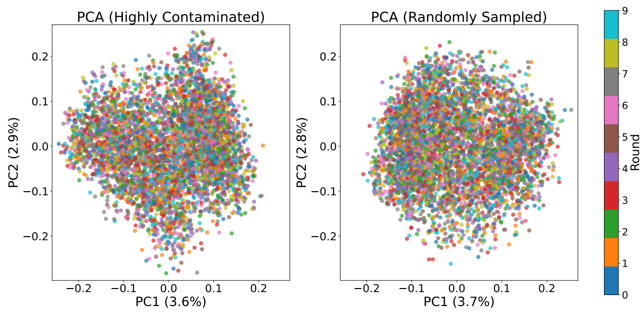

Figure 5: PCA projections of ℓ_2-normalized embeddings for Highly Contaminated (left) and Randomly Sampled (right) splits. Points are colored by round; both splits show full intermixing with PC1/PC2 each explaining $<4\%$ variance.

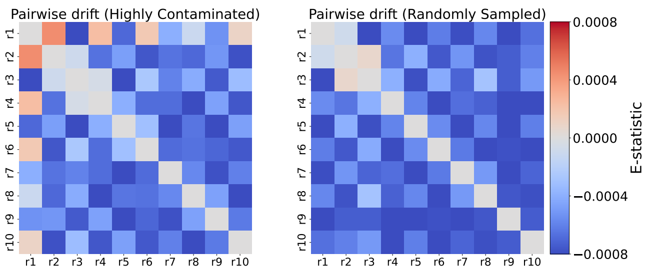

Figure 6: Pairwise drift heatmaps. Left: Highly Contaminated; Right: Randomly Sampled. Cells show the *E-statistic* from the Energy two-sample test (`hyppo.ksample.Energy`) computed on Euclidean distances of ℓ_2-normalized embeddings; 10k permutations with BH–FDR over 45 round pairs.

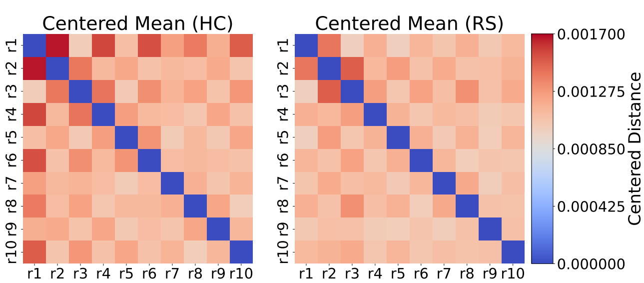

Figure 7: Centered mean distance across rounds. Left: Highly Contaminated; Right: Randomly Sampled. Each cell shows cross-round mean distance minus the average within-round distance (proportional to the biased Energy distance). Values cluster near zero with maxima $\leq 1.7\times10^{-3}$.

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Appendix

Claim Extraction Prompt

You are given a passage of text. Your task is to extract one-sentence standalone factual claims from the passage.

Each claim must:

- Be self-contained (i.e., understandable on its own without requiring coreference resolution)
- Not rely on ambiguous references like “this”, “that”, “they”, etc.
- Be phrased as a complete declarative sentence
- Be verifiable (e.g., “Riot Games has no plans for League of Legends 2.” is acceptable; “They have no plans for League of Legends 2” is not)

For each claim, also provide the exact verbatim supporting text span from the original text that led you to extract the claim.

Output JSON Keys:

- `"claim1"`: standalone claim
- `"supporting_text_span1"`: verbatim supporting text span from original
- `"claim2"`: standalone claim
- `"supporting_text_span2"`: verbatim supporting text span from original
- `...`

Task:
Here is the input text: `{text}`

Temporal-Sequence Multi-Hop QA Prompt

You are generating `{n_pairs}` **temporal-sequence** multi-hop QA pairs.
Buckets of claims (grouped by doc_id) are below:
`{json.dumps(buckets, indent=2)}`

Rules for each QA pair:

- Combine claims from at least **{min_buckets} different buckets**.
- Each chosen claim must mention a year, date, or explicit timeframe.
- Ask for an ordering or an interval (e.g., “How many years after …?”).
- Every referenced fact must be essential for the answer.
- The question should not contain the answer or steps to get the answer.
- The question should not refer to the documents, they should be general.
- List every claim you used by its doc_id and claim_id.
- Provide one concise answer (year, number of years, earlier event, etc.).

Output format (JSON list):

- `"used_claims"`: list of {{doc_id, claim_id, claim}}
- `"question"`: string
- `"answer"`: string

Return only the JSON—no extra commentary.

Comparison-Style Multi-Hop QA Prompt

You are generating `{n_pairs}` **comparison-style** multi-hop QA pairs.
Buckets of claims (grouped by doc_id) are below:
`{json.dumps(buckets, indent=2)}`

Rules for each QA pair:

- Use claims from at least **{min_buckets} different buckets**.
- Frame the question so the solver must contrast values (higher, lower, difference, ratio, earlier, later).
- Keep wording tight—only include values essential for the comparison.
- The question should not contain the answer or steps to get the answer.
- The question should not refer to the documents, they should be general.
- List every claim you used by its doc_id and claim_id.
- Provide a single, concise answer (number, entity, etc.).

Output format (JSON list):

- `"used_claims"`: list of {{doc_id, claim_id, claim}}
- `"question"`: string
- `"answer"`: string

Return only the JSON—no extra commentary.

Conjunction-Style Multi-Hop QA Prompt

You are generating `{n_pairs}` **conjunction-style** multi-hop QA pairs.
Buckets of claims (grouped by doc_id) are below:
`{json.dumps(buckets, indent=2)}`

Rules for each QA pair:

- Combine facts from at least **{min_buckets} different buckets** (can use more).
- The answer must change if any referenced claim were false (logical AND).
- No trivia—every fact must be essential.
- The question should not contain the answer or steps to get the answer.
- The question should not refer to the documents, they should be general.
- List every claim you used by its doc_id and claim_id.
- Provide a single, concise answer (entity, number, date, etc.).

Output format (JSON list):

- `"used_claims"`: list of {{doc_id, claim_id, claim}}
- `"question"`: string
- `"answer"`: string

Return only the JSON—no extra commentary.

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